



**ALLIANCE
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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Sales Performance Analysis Using Power BI

Submitted By: Mahir Mavani

Name: Mahir Mavani

Reg. No.: 2411021240003

Date: 31-08-2026

GitHub Link: [Mahir GitHub Link](#)

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 1

Sales Performance Analysis Using Power BI

Problem Statement

Management wants to quickly understand the sales performance of the business across different products, categories, and regions. The objective of this project is to import the retail sales dataset into Power BI, analyze the data, and create interactive visualizations that help identify important business patterns.

The Sample Superstore dataset was used for the analysis. The dashboard focuses on sales, profit, orders, quantity, product performance, category performance, regional performance, and sales trends over time.

Methodology / Steps Followed

Step 1: Data Collection

- The Sample Superstore dataset was selected for the project.
- The dataset was available in CSV format.
- The dataset contains information related to orders, products, categories, sales, profit, quantity, regions, and order dates.
- The dataset was selected because it provides sufficient information to analyze sales performance from different business perspectives.

Step 2: Data Import

- Opened Power BI Desktop.
- Selected the Get Data option.
- Selected Text/CSV as the data source.
- Imported the Sample Superstore CSV dataset into Power BI.
- Loaded the dataset into Power BI for further analysis.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Segment	Country	City	State	Region	Product ID	Category	Sub-Category
65	CA-2015-135545	24 November 2015	30 November 2015	Standard Class	KM-16720	Consumer	United States	Los Angeles	California	West	OFF-PA-10003892	Office Supplies	Paper
161	CA-2016-162733	11 May 2016	12 May 2016	First Class	TT-21070	Consumer	United States	Los Angeles	California	West	OFF-PA-10002751	Office Supplies	Paper
172	CA-2014-118962	05 August 2014	09 August 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10000659	Office Supplies	Paper
173	CA-2014-118962	05 August 2014	09 August 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10001144	Office Supplies	Paper
559	CA-2016-113243	10 June 2016	15 June 2016	Standard Class	OT-18730	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper
694	CA-2015-131534	28 March 2015	02 April 2015	Standard Class	AB-10165	Consumer	United States	Los Angeles	California	West	OFF-PA-10000743	Office Supplies	Paper
761	CA-2015-112319	31 August 2015	05 September 2015	Standard Class	AR-10510	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper
904	CA-2016-130477	08 April 2016	12 April 2016	Standard Class	LC-17140	Consumer	United States	Los Angeles	California	West	OFF-PA-10002947	Office Supplies	Paper
906	CA-2016-130477	08 April 2016	12 April 2016	Standard Class	LC-17140	Consumer	United States	Los Angeles	California	West	OFF-PA-10000019	Office Supplies	Paper
1144	CA-2014-145969	29 September 2014	03 October 2014	Standard Class	AP-10915	Consumer	United States	Los Angeles	California	West	OFF-PA-10002105	Office Supplies	Paper
1569	US-2017-152002	11 June 2017	11 June 2017	Same Day	DV-13465	Consumer	United States	Los Angeles	California	West	OFF-PA-10000357	Office Supplies	Paper
1594	CA-2014-117317	19 October 2014	19 October 2014	Same Day	JF-15490	Consumer	United States	Los Angeles	California	West	OFF-PA-10004519	Office Supplies	Paper
1609	US-2016-115819	19 April 2016	24 April 2016	Second Class	JO-15280	Consumer	United States	Los Angeles	California	West	OFF-PA-10002377	Office Supplies	Paper
1683	US-2016-101497	29 September 2016	01 October 2016	Second Class	PS-18760	Consumer	United States	Los Angeles	California	West	OFF-PA-10000176	Office Supplies	Paper
1778	CA-2015-105347	24 November 2015	28 November 2015	Standard Class	DP-13000	Consumer	United States	Los Angeles	California	West	OFF-PA-10000675	Office Supplies	Paper
2083	CA-2017-104647	23 February 2017	01 March 2017	Standard Class	CK-12595	Consumer	United States	Los Angeles	California	West	OFF-PA-10002870	Office Supplies	Paper
2248	CA-2015-116092	15 February 2015	18 February 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10004285	Office Supplies	Paper
2252	CA-2015-116092	15 February 2015	18 February 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10000477	Office Supplies	Paper
2413	CA-2014-151295	12 November 2014	16 November 2014	Standard Class	JA-15970	Consumer	United States	Los Angeles	California	West	OFF-PA-10001947	Office Supplies	Paper
2492	CA-2014-113579	13 December 2014	15 December 2014	Second Class	KD-16345	Consumer	United States	Los Angeles	California	West	OFF-PA-10001457	Office Supplies	Paper
2837	CA-2017-110842	11 November 2017	16 November 2017	Standard Class	GA-14725	Consumer	United States	Los Angeles	California	West	OFF-PA-10000520	Office Supplies	Paper
2888	CA-2017-102099	18 December 2017	19 December 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10003172	Office Supplies	Paper
2889	CA-2017-102099	18 December 2017	19 December 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10000289	Office Supplies	Paper
3422	CA-2016-134803	28 May 2016	03 June 2016	Standard Class	CL-12565	Consumer	United States	Los Angeles	California	West	OFF-PA-10003790	Office Supplies	Paper
3557	CA-2017-118542	01 December 2017	05 December 2017	Standard Class	CC-12550	Consumer	United States	Los Angeles	California	West	OFF-PA-10004735	Office Supplies	Paper
3565	CA-2016-130029	03 July 2016	06 July 2016	First Class	GT-14755	Consumer	United States	Los Angeles	California	West	OFF-PA-10000552	Office Supplies	Paper
3945	CA-2016-133144	16 May 2016	21 May 2016	Second Class	DO-13435	Consumer	United States	Los Angeles	California	West	OFF-PA-10004971	Office Supplies	Paper
3971	CA-2014-135090	22 November 2014	26 November 2014	Standard Class	SP-20920	Consumer	United States	Los Angeles	California	West	OFF-PA-10002245	Office Supplies	Paper
3985	CA-2016-135265	07 July 2016	09 July 2016	Second Class	CC-12370	Consumer	United States	Los Angeles	California	West	OFF-PA-10002923	Office Supplies	Paper
4118	CA-2015-153416	24 November 2015	29 November 2015	Standard Class	TS-21340	Consumer	United States	Los Angeles	California	West	OFF-PA-10002005	Office Supplies	Paper

Step 3: Data Preparation

- Checked the available columns and data types in the dataset.
- Verified fields such as Sales, Profit, Quantity, Order ID, Order Date, Category, Sub-Category, Product Name, and Region.
- Ensured that the numerical fields were suitable for calculations.
- Ensured that the Order Date field could be used for time-based analysis.
- The data was then prepared for visualization and analysis.

Step 4: Creating DAX Measures

Several measures were created in Power BI to calculate important business metrics.

Total Sales

```
1 Total Sales = SUM('Sample_ Superstore'[Sales])
```

Total Profit

```
1 Total Profit = SUM('Sample_ Superstore'[Profit])
```

Total Quantity

```
1 Total Quantity = SUM('Sample_ Superstore'[Quantity])
```

Total Orders

```
1 Total Orders = DISTINCTCOUNT('Sample_ Superstore'[Order ID])
```

Profit Margin

```
1 Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
```

Step 5: Dashboard Development

Two interactive report pages were developed in Power BI.

Page 1 – Sales Performance Dashboard

The first page provides an overall view of business performance.

KPI Cards

The following KPI cards were created:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Profit Margin

Visualizations

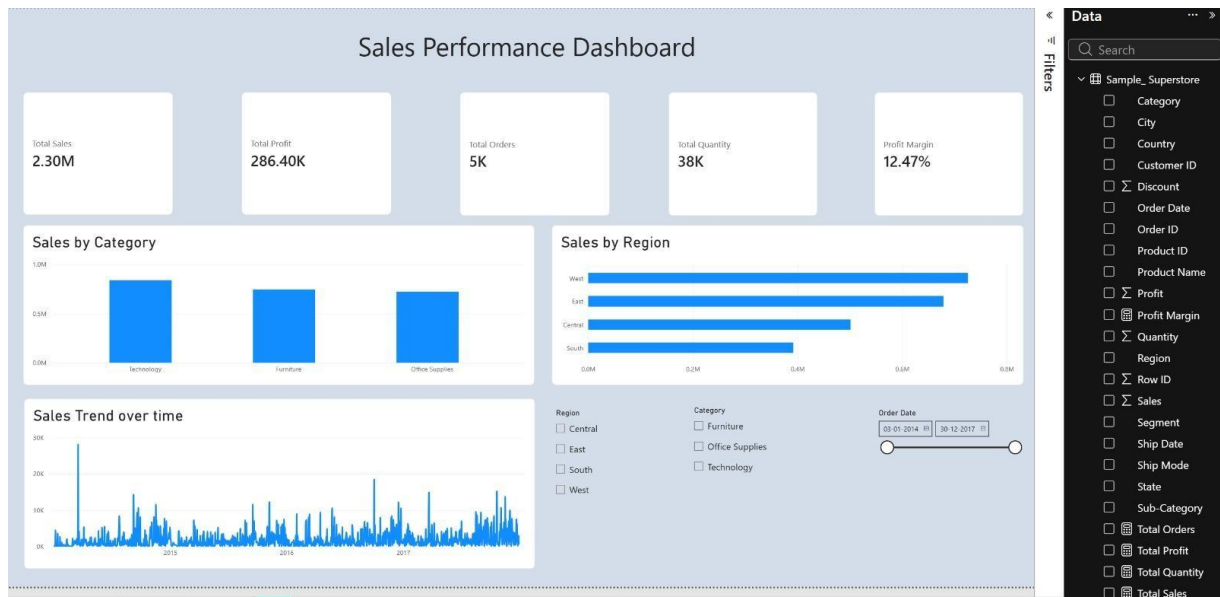
The following charts were created:

1. **Sales by Category**
 - Compares sales across Furniture, Office Supplies, and Technology.
2. **Sales by Region**
 - Compares sales performance across Central, East, South, and West regions.
3. **Sales Trend over Time**
 - Shows how sales changed throughout the dataset's time period.

Interactive Filters

Slicers were added for:

- Category
- Region
- Order Date



Page 2 – Product & Category Analysis

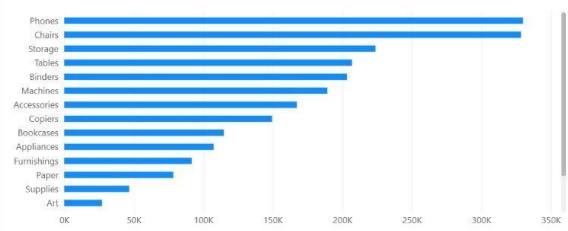
The second page focuses on product-level and sub-category-level performance.

Visualizations

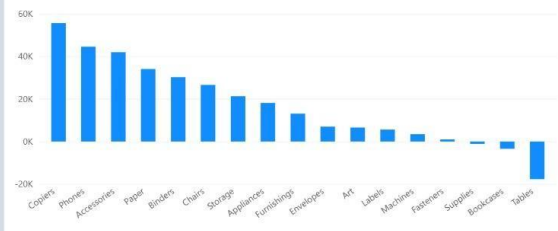
1. **Sales by Sub-Category**
 - Shows the sales generated by different product sub-categories.
2. **Profit by Sub-Category**
 - Compares profitability across different sub-categories.
3. **Top 10 Products by Sales**
 - Identifies the ten products with the highest sales.
4. **Top 10 Products by Profit**
 - Identifies the ten products generating the highest profit.

Product & Category Analysis

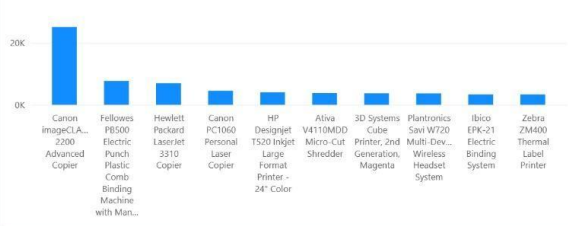
Sales by Sub-Category



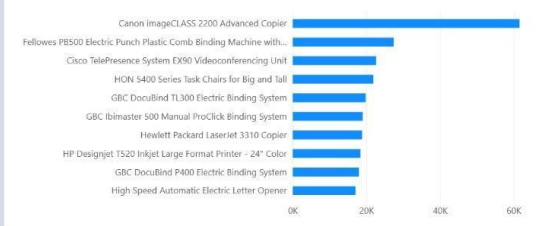
Profit by Sub-Category



Top 10 Products by Profit



Top 10 Products by Sales



Output

The final Power BI report provides an interactive view of retail business performance.

The dashboard provides:

- Overall Sales Performance
- Overall Profit Performance
- Total Order and Quantity Analysis
- Category-wise Sales Analysis
- Region-wise Sales Analysis
- Sales Trends over Time
- Sub-Category Sales Analysis
- Sub-Category Profit Analysis
- Top 10 Products by Sales
- Top 10 Products by Profit
- Interactive Filters and Slicers

The visualizations help management quickly identify high-performing categories, products, and regions and also identify areas that may require attention.

Key Business Insights

Based on the visualizations created in Power BI, the following patterns were identified:

1. Technology is the highest-selling category, followed by Furniture and Office Supplies.
2. The West region records the highest sales, while the South region has the lowest sales among the four regions.
3. Phones and Chairs are among the leading sub-categories in terms of sales.
4. Copiers are the most profitable sub-category based on the profit analysis.
5. Tables show negative profit, even though they generate considerable sales. This indicates that this sub-category may require further investigation regarding pricing, discounts, or costs.
6. The Top 10 products by sales and Top 10 products by profit are not exactly the same, showing that high sales do not always result in high profitability.
7. The sales trend visualization shows variations in sales over time, helping management understand changes in business performance.

Conclusion

The project successfully used Power BI to analyze the sales performance of the Sample Superstore dataset. Different DAX measures and interactive visualizations were created to provide a clear understanding of sales, profit, products, categories, and regions.

The developed dashboard allows management to quickly compare business performance and identify important patterns such as high-performing categories, profitable sub-categories, top products, and regional differences. The analysis demonstrates how Power BI can transform raw sales data into meaningful information that can support data-driven business decisions.

Learning Outcomes

Through this project, I learned:

- How to import a CSV dataset into Power BI.
- How to examine and prepare data for analysis.
- How to create DAX measures for business calculations.
- How to calculate Total Sales, Total Profit, Total Orders, Total Quantity, and Profit Margin.
- How to create KPI cards in Power BI.
- How to create bar charts, column charts, and line charts.
- How to analyze sales across categories, sub-categories, products, and regions.

- How to apply Top N filters to identify the top-performing products.
- How to create interactive slicers and filters.
- How to identify useful business patterns from visualizations.
- How to design an interactive Power BI dashboard for management reporting.